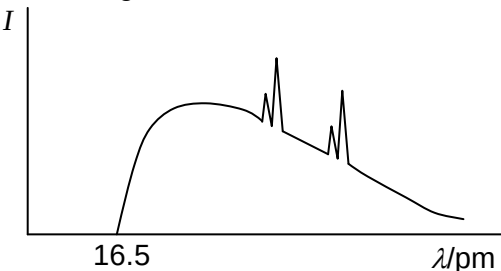


Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
7	(a)	(i)	Background spectra as skewed normal curve and line spectra shown (1) Minimum wavelength labelled /shown as 1.65×10^{-11} m (1) e.g. 	1	1		2	1	
		(ii)	Power = 9 000 W (1) Rate of heat = 8 937 W (1)		2		2	2	
		(iii)	Rearrangement $\mu = \frac{\ln\left(\frac{100}{60}\right)}{4.1 \times 10^{-3}}$ OR 0.6×0.6 (1) $\mu = 365 \text{ m}^{-1}$ answer with unit OR 0.36 (1) Substitution into $I = I_0 e^{-\mu x}$ where $x = 2.8 \times 10^{-3}$ so $I = 36\%$ (1)		3		3	3	
	(b)	(i)	Ultrasound hits blood cells (1) Reflected pulse different frequency/wavelength (1)	2			2		
		(ii)	Rearrangement $v = \frac{\Delta f c}{2 f_0 \cos \theta}$ (1) $v = 0.098 \text{ [m/s]}$ (1) Answer of 0.18 [m/s] award 1 mark only for the question part		2		2	2	